

Regime-Based World Modeling: A Layered Epistemic Architecture for Complex Systems

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Abstract

Modeling complex systems requires navigating phenomena that exhibit distinct behavioral rules at different scales of organization. A persistent challenge in multi-scale modeling is *regime blindness*—the implicit assumption that variables and laws valid in one regime remain primary in another. This paper introduces a **Regime-Based World Modeling (RBWM)** framework, a meta-epistemic architecture designed to map model validity across two orthogonal axes: the **Regime Axis** (R), representing emergent complexity levels, and the **Epistemic Layer Axis** (L), representing the derivation distance from raw observation to theoretical construct.

We generalize the **Regime-Gating Principle**, positing that a scientific claim is only epistemically valid if its primary variables are “driver-near” within the phenomenon’s operative regime. By assigning every model component a coordinate pair (R, L) , the framework resolves apparent paradoxes between conflicting models as non-contradictory descriptions occupying different epistemic coordinates. We demonstrate that errors often arise not from incorrect data, but from *regime leakage*—the application of high-level theoretical constructs as proxies for low-level drivers outside their native validity domain.

Keywords: Emergence, Multi-scale Modeling, Epistemic Layering, Regime-Gating, Driver Proximity, Complexity Architecture, World Models.

1. Introduction

1.1 The Challenge of Multi-Scale Consistency

The central problem in modeling reality is not the scarcity of data, but the *incoherence of descriptions across scales*. A single physical system can be described as a quantum field, a thermodynamic ensemble, a fluid, or a solid object depending on the scale of observation. While reductionism seeks to derive all behaviors from the lowest level, and holism emphasizes the emergent whole, neither approach provides a rigorous protocol for how to *transition* between these descriptions without losing epistemic integrity.

Current world models—whether in computational physics, climate science, or artificial intelligence—often suffer from what we term **regime blindness**. They frequently conflate variables that are *causal drivers* in one regime with variables that are merely *statistical proxies* in another. This methodological ambiguity leads to “regime leakage,” where models calibrated for one level of complexity are invalidly extrapolated to another, generating phantom anomalies and theoretical dead-ends.

1.2 A Two-Axis Epistemic Architecture

To address this, we propose a formalized meta-framework, **Regime-Based World Modeling (RBWM)**, which organizes scientific claims along two orthogonal axes:

The Regime Axis (R): Defines the level of emergent complexity. We distinguish three primary regimes:

- R_0 (**Field/Statistical**): Phenomena governed by statistical mechanics and field theories—CMB fluctuations, quantum states, thermodynamic ensembles.
- R_1 (**Structure/Relational**): Phenomena where stable structures emerge from field dynamics—galaxies, ecosystems, neural networks, social institutions.
- R_2 (**Complex Adaptive**): Phenomena exhibiting feedback, learning, and self-modification—economies, cognition, AI systems, evolutionary processes.

The Epistemic Layer Axis (L): Defines the assumption load of a variable, ranging from direct observation to theoretical construction:

- L_0 (**Raw Observation**): Unprocessed sensor data, photon counts, voltage measurements.
- L_1 (**Calibrated Observable**): Instrument-corrected quantities—calibrated magnitudes, corrected spectra, standardized metrics.
- L_2 (**Regime-Near Physics**): Derived quantities proximal to the causal drivers in the operative regime—acceleration, temperature gradients, population dynamics.
- L_3 (**Theoretical Construct**): Model-dependent entities—dark matter halos, utility functions, attention mechanisms.

1.3 The Problem This Solves

Consider a concrete example: the “missing mass” problem in galaxy dynamics. In the standard approach, rotation curves are measured (L_1), fitted to Newtonian predictions (L_3), and the discrepancy is attributed to “dark matter” (L_3). This treats an L_3 construct as if it were an L_0 observable. The RBWM framework makes this category error *visible* by requiring explicit (R, L) coordinates for all claims.

By contrast, treating the dynamical response to *gradients* (L_2) as the primary variable in the structural regime (R_1) may yield different—and potentially more parsimonious—explanations. The framework does not adjudicate which physics is correct; it reveals which claims are *epistemically comparable* and which are not.

2. The Regime-Gating Principle

2.1 Core Formulation

The central contribution of this paper is the generalization of the **Regime-Gating Principle (RGP)**:

A scientific claim C operating on variable V is epistemically valid only if V is “driver-proximal” within the phenomenon’s operative regime R .

We define **driver proximity** as follows: A variable V has high driver proximity in regime R if small changes in V produce first-order effects on the system’s dynamics *within that regime*. By contrast, a variable with low driver proximity is a *statistical consequence* of the actual drivers—it correlates with outcomes but does not cause them in the relevant sense.

2.2 Why This Matters

Regime-gating prevents a class of errors we call **proxy chain collapse**. In complex systems, long inferential chains connect observations (L_0) to conclusions (L_3). Each link in the chain introduces assumptions. Without explicit tracking, scientists may inadvertently treat a variable that is three derivational steps removed from observation *as if* it were directly observed.

The **Regime-Consistent Measurement Principle (RCMP)**, a specialized application of RGP, states:

“Interpret observables through variables nearest the physical driver in the operative regime.”

This rule prevents L_3 constructs from being treated as L_0 data, a common error in fields ranging from cosmology to economics.

2.3 Regime Leakage

When RGP is violated, **regime leakage** occurs. This manifests as:

- **Phantom anomalies:** Discrepancies between model and data that arise from applying the wrong regime’s variables.
- **Theoretical proliferation:** Invention of new entities (dark matter, dark energy, hidden layers) to explain discrepancies that are actually category errors.
- **Cross-domain incompatibility:** Models that work in one regime but fail catastrophically in another.

3. The (R, L) Coordinate System

3.1 Mapping Claims

Every scientific claim can be assigned a coordinate pair (R, L) . This assignment is not arbitrary but follows from analysis of:

1. **Scale of the phenomenon:** What level of emergent complexity does the claim address?
2. **Derivation chain:** How many inferential steps separate the claim from raw observation?
3. **Driver proximity:** Does the variable in question directly cause effects in this regime?

3.2 Resolving Model Conflicts

A key application of the (R, L) system is *conflict resolution*. Consider two competing theories:

- **Theory A** (Λ CDM in cosmology): “Dark matter halos cause flat rotation curves.” — Coordinates: $(R_0 \rightarrow R_1, L_3)$
- **Theory B** (MOND/RAR): “Modified dynamics in low-acceleration regimes explain rotation curves.” — Coordinates: (R_1, L_2)

These theories appear contradictory, but the (R, L) system reveals they operate at different coordinates. Theory A imports an R_0 field-theoretic construct (dark matter) into R_1 phenomenology. Theory B stays within R_1 and uses driver-near variables (acceleration). They are not testing the same hypothesis at the same coordinates—they are *different descriptions of different aspects* of the same system.

This does not mean both are equally correct. It means *comparison requires coordinate matching*. A fair test must compare claims at the same (R, L) coordinates.

3.3 The Coordinate Validity Domain

Each model has a **validity domain**: the range of (R, L) coordinates where its primary variables remain driver-proximal. Outside this domain, the model may still correlate with data, but it loses *explanatory power*—it becomes a fit rather than a theory.

Formally, we define:

$$\mathcal{V}(M) = \{(R, L) : \text{variables of } M \text{ are driver-proximal at } (R, L)\} \quad (1)$$

The validity domain $\mathcal{V}(M)$ is the set of coordinates where model M provides mechanistic (not just statistical) explanations. Claims outside $\mathcal{V}(M)$ require explicit disclaimers about their epistemic status.

4. Applications Across Domains

4.1 Physics: From Particles to Cosmology

Physics provides the clearest examples of regime structure:

- **Quantum mechanics** (R_0, L_2-L_3) : Wave functions are theoretical constructs; probabilities are derived from measurements.
- **Statistical mechanics** $(R_0 \rightarrow R_1, L_2)$: Temperature and pressure emerge as statistical properties of microscopic dynamics.
- **Astrophysics** (R_1, L_1-L_3) : Luminosity distances (L_1) , Hubble parameter (L_2) , dark energy equation of state (L_3) .

The RBWM framework clarifies why “quantum gravity” remains elusive: it attempts to merge descriptions from incompatible (R, L) coordinates without an explicit bridging protocol.

4.2 Biology: Cells to Organisms

Biological systems exhibit clear regime stratification:

- **Molecular** (R_0) : Chemical kinetics, gene expression rates.
- **Cellular** (R_1) : Organelle function, metabolic pathways, signaling networks.
- **Organismal** (R_2) : Behavior, homeostasis, adaptation.

The “genotype-phenotype gap” is a manifestation of regime leakage: attempting to explain R_2 phenomena (behavior) directly from R_0 data (gene sequences) without accounting for the intermediate regime structure.

4.3 Economics: Agents to Markets

Economic modeling suffers extensively from regime blindness:

- **Microeconomics** (R_1) : Individual agents, utility maximization, local equilibria.
- **Macroeconomics** (R_2) : Aggregate demand, GDP, inflation, systemic risk.

The 2008 financial crisis exposed regime leakage: models calibrated on R_1 dynamics (individual risk) failed catastrophically when applied to R_2 phenomena (systemic contagion). The (R, L) framework would have flagged the mismatch.

4.4 Artificial Intelligence: Parameters to Behavior

AI systems present a novel application:

- **Parameter space** (R_0, L_3): Weights, architectures, training dynamics.
- **Representation space** (R_1, L_2): Embeddings, attention patterns, activations.
- **Behavioral space** (R_2, L_1): Outputs, decisions, emergent capabilities.

“Mechanistic interpretability” research is an implicit acknowledgment of regime structure: understanding R_2 behavior requires tracing causal chains through R_1 representations, not just correlating with R_0 parameters.

5. Implementation in World Models

5.1 Coordinate Tagging Protocol

We propose a practical protocol for implementing RBWM:

1. **Explicit tagging:** Every model component receives an (R, L) coordinate label.
2. **Transition functions:** When crossing regime boundaries, explicit coarse-graining or emergence operators are specified.
3. **Validity checking:** Before combining claims from different coordinates, check whether bridging is epistemically licensed.
4. **Uncertainty propagation:** Track how epistemic uncertainty increases with derivational distance (L) and regime transitions (R).

5.2 Application to AI World Models

For artificial intelligence systems building internal world models, RBWM provides:

- **Regime-aware inference:** The system knows which variables are driver-proximal in which contexts.
- **Transparent reasoning:** (R, L) coordinates make the system’s assumptions explicit and auditable.
- **Graceful degradation:** When operating outside a model’s validity domain, the system can flag uncertainty rather than extrapolate confidently.

This architecture is particularly relevant for autonomous scientific agents that must reason across multiple domains without human oversight.

5.3 The Symbiose Implementation

The concepts presented here have been implemented in a working knowledge architecture called **Symbiose**, which integrates:

- **Graph structures (Neo4j):** Symbolic relationships between concepts, explicit regime tagging.
- **Vector embeddings (Qdrant):** Semantic similarity search, long-term memory retrieval.
- **Relational learning (GNN):** Discovery of latent structure across regime boundaries.
- **Metakognition:** Self-monitoring, epistemic surprise detection, why-chain tracing.

The system operationalizes the RBWM framework, demonstrating that the architecture is not merely theoretical but *computationally realizable*.

6. Discussion

6.1 What This Framework Is Not

RBWM is *not* a physical theory. It does not claim that the universe is structured in a particular way. It is an **epistemic architecture**—a set of principles for organizing knowledge *about* complex systems, whatever their underlying nature.

The framework is also not relativistic in the sense of claiming all models are equally valid. Some models have larger validity domains than others. Some are more parsimonious. The (R, L) system provides tools for *making these comparisons rigorous*.

6.2 Relation to Existing Frameworks

RBWM builds on and generalizes several existing approaches:

- **Effective field theory** in physics: EFT already acknowledges that descriptions are valid only within certain energy scales. RBWM generalizes this to any scale and any domain.
- **Levels of organization** in biology: The hierarchical view of life (molecule $\rightarrow$ cell $\rightarrow$ organism) is a precursor to regime stratification.
- **Epistemic layering** in philosophy of science: The distinction between observable and theoretical terms maps onto the L -axis.

6.3 Limitations

Several limitations should be acknowledged:

1. **Coordinate assignment is non-trivial:** Determining the (R, L) coordinates of a claim requires domain expertise and may be contested.
2. **Regime boundaries are fuzzy:** The transition from R_0 to R_1 is not sharp; phase-extended transitions may be required.
3. **Not algorithmically complete:** The framework provides principles, not an automatic procedure for resolving all disputes.

7. Conclusion

We have presented **Regime-Based World Modeling (RBWM)**, a meta-epistemic architecture for navigating multi-scale complexity. The framework’s core contributions are:

1. **The (R, L) coordinate system:** A two-axis mapping of regime (emergent complexity level) and epistemic layer (derivational distance from observation).
2. **The Regime-Gating Principle:** Claims are valid only where their variables are driver-proximal.
3. **Regime leakage detection:** A diagnostic for identifying category errors in multi-scale modeling.
4. **Conflict resolution protocol:** Competing models are compared at matched (R, L) coordinates.

The framework has been demonstrated across physics, biology, economics, and artificial intelligence, and has been implemented in a working computational system (Symbiose). We

propose that RBWM offers a robust, transparent, and regime-aware foundation for building world models capable of navigating the complexity of natural and artificial systems.

In essence: *The map is not the territory—but with proper coordinates, we can at least know which territory we are mapping.*

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